

AMENDMENT NO. 3 FEBRUARY 2009
TO
IS 3513 (PART 1) : 1989 SPECIFICATION FOR RESIN
TREATED COMPRESSED WOOD LAMINATES
(COMPREGS)

PART 1 FOR ELECTRICAL PURPOSES

(First Revision)

(Page 2, clause 5.2) — Substitute the following for the existing:

‘The synthetic resins used for impregnation and bonding of veneers shall be of phenol or cresol formaldehyde type and the resultant resin adhesive shall conform to the requirements specified in IS 848 : 2006.’

(Page 4, Table 1) — Substitute the following for the existing table:

Table 1 Physical Properties of Compregs (All Types)
(Clause 8.1)

Sl No.	Property	Requirement		Method of Test
		Types I, II, III and IV	Types V and VI	
i)	Specific gravity, <i>Min</i>	1.25	1.25	IS 1708 (Part 2) : 1986
ii)	Moisture content and volatile matter, percent, <i>Max</i>	4	4	IS 1708 (Part 1) : 1986
iii)	Water absorption at $27 \pm 2^\circ\text{C}$, percent, <i>Max</i>	1.2	1.5	IS 3513 (Part 4) : 1966
iv)	Sporadic working temperature, <i>Max</i>	90°C	90°C	IS 3513 (Part 4) : 1966

[Page 5, Table 3, (see also Amendment No. 1)] — Substitute the following for the existing table:

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**Table 3 Mechanical Properties
(Clause 8.3)**

Sl No.	Test	Unit	Requirement		Method of Test
			Type V	Type VI	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Tensile strength, <i>Min</i> :				
	a) Along the grains	MPa	140	70	IS 1734 (Part 9) : 1983
	b) Across the grains	MPa	—	50	
ii)	Static bending strength, <i>Min</i> :				
	a) Along the grains	MPa	185	90	IS 1998 : 1962
	b) Across the grains	MPa	—	65	
iii)	Compressive strength, <i>Min</i> :				
	a) Parallel to laminae	MPa	160	110	IS 1708 (Parts 8 & 9) : 1986
	b) Perpendicular to laminae	MPa	—	170	
iv)	Shear strength, <i>Min</i> :				
	a) Parallel to grains & laminae	MPa	12	15	IS 1708 (Part 11) : 1986
	b) Perpendicular to grains & laminae	MPa	50	40	
v)	Hardness, Rockwell, <i>Min</i> , M scale		70	70	IS 1586 : 2000 ¹⁾
vi)	Impact strength, Izod, Un-notched, <i>Min</i> :				
	a) Perpendicular to laminae	kg/m ²	0.45	0.30	IS 1998 : 1962
	b) Parallel to laminae	kg/m ²	0.25	0.20	

¹⁾ Rockwell hardness in M scale shall be determined using an indenter of diameter of 6.350 ± 0.0025 mm and with Minor load of 10 kg and Major load of 100 kg.

(Page 6, Annex A) — Substitute the following for the existing entries:

'IS 848 : 2006	Synthetic resin adhesives for plywood (phenolic and aminoplastic) — Specification (<i>second revision</i>)
IS 1586 : 2000	Method for Rockwell hardness test for metallic material (scales A-B-C-D-E-F-G-H-K 15N, 30N, 45N, 15T, 30T and 45T) (<i>third revision</i>)'

(CED 20)

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